

# Multi-Modal, Training-Free Crack Extraction via Generalized FRANGI Graph

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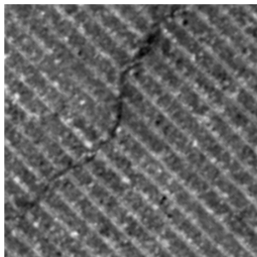
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<sup>2</sup>Cerema, ENDSUM research team, France

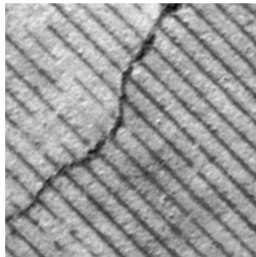
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# Problem setting



Intensity [FIND 22]



Range / depth [FIND 22]

- ▶ Cracks are **thin, weak** and often **discontinuous**
- ▶ Several **image modalities** may observe the same geometry
- ▶ Goal: an **editable crack skeleton, without training** [EUVIP 26]

[FIND 22] S. Zhou et al., "Fused Image dataset for convolutional neural Network-based crack Detection (FIND)." Zenodo, 2022.

[EUVIP 26] L. Hauseux et al., "Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph." EUVIP, 2026.

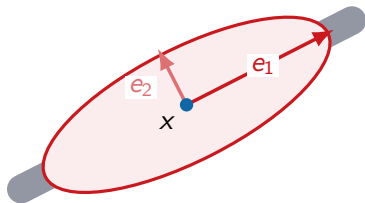
# 01

## From pixels to a FRANGI graph



# Classical FRANGI filter

$$\mathcal{H}_\sigma(x) = \nabla^2(G_\sigma * I)(x), \quad |\lambda_1| \leq |\lambda_2|$$



Local ridge geometry [Frangi 98]

► **Shape:**  $R_B = |\lambda_1|/|\lambda_2|$

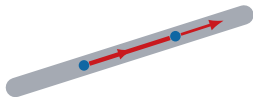
► **Contrast:**  $S = \sqrt{\lambda_1^2 + \lambda_2^2}$

$$F_\sigma(x) = \exp\left(-\frac{R_B^2}{2\beta^2}\right) \left(1 - \exp\left(-\frac{S^2}{2c^2}\right)\right)$$

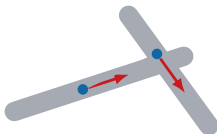


# Our contribution: generalized FRANGI graph

Aligned axes  
Strong edge



Misaligned axes  
Weak edge



Pairwise orientation alignment [GRETSI 25] [EUVIP 26]

- ▶ Pixels become **nodes**
- ▶ Ridge compatibility becomes **edge similarity**
- ▶ **Alignment** refines intensity and shape information:

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} S_{\text{int}}^{\sigma} S_{\text{shape}}^{\sigma} S_{\text{align}}^{\sigma}$$

[GRETSI 25] L. Hauseux et al., "Généralisation du filtre de Frangi pour l'extraction des réseaux de fissures..." GRETSI, 2025.

[EUVIP 26] L. Hauseux et al., "Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph." EUVIP, 2026.



# Multimodal fusion

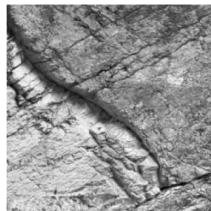
1. Normalize each modality's Hessian:

$$\hat{\mathcal{H}}_{\sigma}^{(m)}(x) = \frac{\mathcal{H}_{\sigma}^{(m)}(x)}{\max_{y \in \Omega} \|\mathcal{H}_{\sigma}^{(m)}(y)\|_2}$$

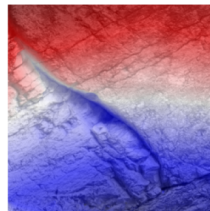
2. Fuse operators before eigendecomposition, with  $w_m \geq 0$  and  $\sum_m w_m = 1$ :

$$\mathcal{H}_{\sigma}^{\text{fused}}(x) = \sum_m w_m \hat{\mathcal{H}}_{\sigma}^{(m)}(x)$$

## Palais des Papes [EUVIP 26]



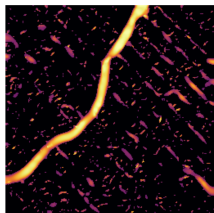
Intensity  
© Cerema



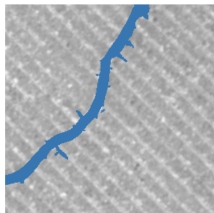
Depth overlay  
© Cerema



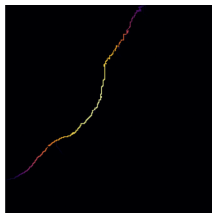
# Extraction pipeline



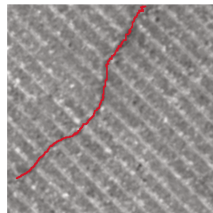
Similarity



Component



Centrality



Skeleton [EUVIP 26]



[EUVIP 26] L. Hauseux et al., "Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph." EUVIP, 2026.

# 02

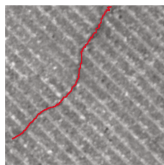
## Experimental results



# FIND: clean data



Ground truth  
[FIND 22]



Our result  
[EUVIP 26]

| Method                         | IoU $\uparrow$ | Tversky $\uparrow$ | Wasserstein $\downarrow$ |
|--------------------------------|----------------|--------------------|--------------------------|
| Our graph [EUVIP 26]           | 63%            | 71%                | 11 px                    |
| CrackSegDiff [CrackSegDiff 24] | <b>87%</b>     | <b>90%</b>         | <b>5 px</b>              |

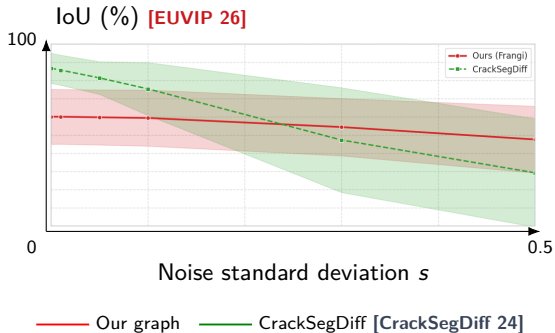
On clean data a trained model scores higher — at the price of supervision, and possibly of overfitting to FIND

[FIND 22] S. Zhou et al., “Fused Image dataset for convolutional neural Network-based crack Detection (FIND).” Zenodo, 2022.

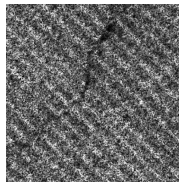
[EUVIP 26] L. Hauseux et al., “Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph.” EUVIP, 2026.

[CrackSegDiff 24] X. Jiang et al., “CrackSegDiff: Diffusion Probability Model-based Multi-modal Crack Segmentation.” arXiv, 2024.

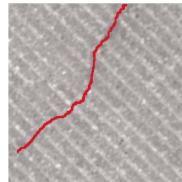
# FIND: synthetic noise



$$I_s = I(1 + N), \quad R_s = R + N'$$
$$N, N' \sim \mathcal{N}(0, s^2) \text{ independent}$$



Noisy intensity



Our result,  $s = 0.5$

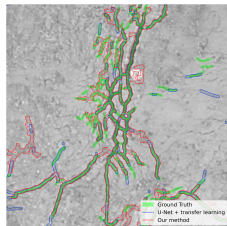
- Slower degradation under added noise

[EUVIP 26] L. Hauseux et al., "Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph." EUVIP, 2026.

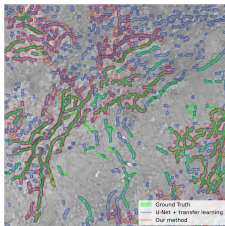
[CrackSegDiff 24] X. Jiang et al., "CrackSegDiff: Diffusion Probability Model-based Multi-modal Crack Segmentation." arXiv, 2024.

# Transfer to geological images

## Vaches Noires [EUVIP 26]



Test 1  
© Cerema

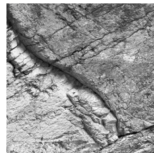


Test 2  
© Cerema

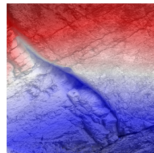
Our graph   Ground truth   U-Net

- IoU 47 / 50% and 29 / 34% (ours / adapted U-Net)

## Palais des Papes [EUVIP 26]



Intensity  
© Cerema



Depth overlay  
© Cerema



Our result  
© Cerema

- Main fracture recovered
- No retraining

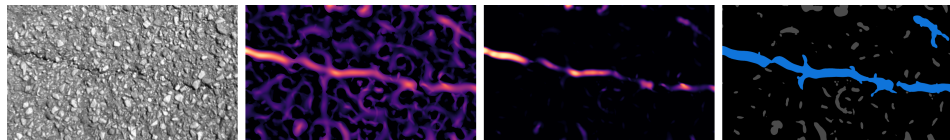
[EUVIP 26] L. Hauseux et al., "Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph." EUVIP, 2026.

# 03

## Beyond pairwise connectivity



# Toward order $K = 2$



Granular image

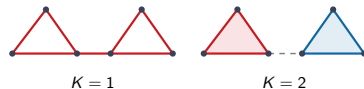
Pixel-wise FRANGI

Graph similarity

Retained component

VT-GraF © Cerema

- ▶ Granular background can create spurious chains
- ▶  $K = 1$ : connectivity through vertices
- ▶  $K = 2$ : connectivity through triangles [ANS 26]



[ANS 26] L. Hauseux et al., "Generalization of Single-Linkage with higher-order interactions." Appl. Netw. Sci., 2026.

# Conclusion & perspective

## Main steps

- ▶ FRANGI filter  $\rightarrow$  FRANGI graph [EUVIP 26]
- ▶ Multimodal fusion at the Hessian level
- ▶ Training-free extraction of crack skeletons

## Next steps

- ▶ Evaluate higher-order connectivity
- ▶ Broaden the benchmarks
- ▶ Can a FRANGI graph hierarchy help guide SAM / CrackSAM?  
[SAM 23] [CrackSAM 24]



Code & data



[EUVIP 26] L. Hauseux et al., "Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph." EUVIP, 2026.

[SAM 23] A. Kirillov et al., "Segment Anything." ICCV, 2023.

[CrackSAM 24] K. Ge et al., "Fine-tuning vision foundation model for crack segmentation in civil infrastructures." Constr. Build. Mater., 2024.

# Bibliography: method and experiments

[Frangi 98]

A. F. Frangi, W. J. Niessen, K. L. Vincken, M. A. Viergever.  
"Multiscale vessel enhancement filtering." MICCAI, 1998.

[GRETSI 25]

L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia.  
"Généralisation du filtre de Frangi pour l'extraction des réseaux de fissures au sein d'images d'un glissement de terrain. Comparaison avec une méthode d'apprentissage profond." GRETSI, 2025.

[EUVIP 26]

L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia.  
"Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph." EUVIP, 2026.

[FIND 22]

S. Zhou, C. Canchila, W. Song.  
"Fused Image dataset for convolutional neural Network-based crack Detection (FIND)." Zenodo, 2022. [zenodo.org/records/6383044](https://zenodo.org/records/6383044).

[CrackSegDiff 24]

X. Jiang, L. Jiang, A. Wang, K. Zhu, Y. Gao.  
"CrackSegDiff: Diffusion Probability Model-based Multi-modal Crack Segmentation." arXiv, 2024. [arXiv:2410.08100](https://arxiv.org/abs/2410.08100).



# Bibliography: higher order and perspective

[ANS 26]

L. Hauseux, K. Avrachenkov, J. Zerubia.  
"Generalization of Single-Linkage with higher-order interactions." *Applied Network Science*, 11, article 9, 2026.

[SAM 23]

A. Kirillov et al.  
"Segment Anything." ICCV, pp. 4015–4026, 2023. [arXiv:2304.02643](https://arxiv.org/abs/2304.02643).

[CrackSAM 24]

K. Ge, C. Wang, Y. Guo, Y. Tang, Z. Hu, H. Chen.  
"Fine-tuning vision foundation model for crack segmentation in civil infrastructures." *Construction and Building Materials*, 431, 136573, 2024.  
[doi.org/10.1016/j.conbuildmat.2024.136573](https://doi.org/10.1016/j.conbuildmat.2024.136573).